A quartic in 204 variables from recovered elliptic differences

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Abstract

Under the stated inherited arithmetic premises, we construct a family of integer quartics in 204 total variables, eight fewer than the 212-variable predecessor. The family is a sum of 186 primitive quadratic squares and has exactly 3046 ordinary monomials. Two elliptic differences are recovered from reciprocal point pairs instead of being stored separately. A positive weighted trace controls the enlarged point domain, including half-integral indices and torsion differences. The old Gaussian congruences also permit the required half-generator in the completeness argument. The coefficient-height improvement is preserved. The sufficient defining parameter remains ineffective.

1 Statement and precise scope

All variable counts include the input. Set

$$K = \mathbb{Q}(\rho), \quad \rho^2 = 2, \quad d = 3 - \rho, \quad D = d^2, \quad E : y^2 = x^3 - Dx,$$

$$R_0 = (1 + \rho, 2\rho), \qquad P = 2R_0.$$

We retain $2E(K) = \mathbb{Z}P$ with P non-torsion, and the precise scalar, pole, contact-height, and fixed-data prime-selection inputs of the 212-variable source [1]. These are implications with explicit arithmetic premises; the finite coefficient checks do not prove them. The full external arithmetic audit is not declared complete. No additional *abc* or Szpiro hypothesis is introduced.

Theorem 1.1. *Under these inputs, there is a computable family*

$$P_N \in \mathbb{Z}[X, Y_1, \ldots, Y_{203}], \qquad N \geq 1,$$

with degree exactly four, integer content one, and exactly 3046 nonzero coordinate monomials. There is one fixed integer N_0 such that every $N \geq N_0$ has rational input projection exactly $\mathbb{Z}$. No numerical sufficient N_0 is determined.

For the balanced presentation put $C = 149^N$, $K_0 = 31581660^{28}$, $J_0 = 5898050446009651968$, and $H_0 = (J_0K_0)^2$. Its exact coefficient height, for every $N \geq 1$, is

$$H(P_N) = \max(H_0, K_0C^2).$$

It equals $31581660^{28}149^{2N}$ for $N \geq 57$. This is a coefficient threshold, not a sufficient defining threshold.

The complete quadratic and expanded quartic tables, their producer, and the independent reconstruction checker are in

`computation/quartic/reciprocal204_v27`.

They specify every coefficient, not just a variable count or an unexpanded expression. The arithmetic proof is separate below.

2 Retained equations and arithmetic conclusions

Start with the unbalanced 212-coordinate table, SHA-256

b2f32d64ef1fb2591d37cfd41aadf8869ce691c39ed92e0e597ec10aaa53fc75.

All indices in the transformation below refer to that table before compaction. Write $R = x_{118}$ and $R_2 = x_{117}$. The retained scalar rows, reference-point rows and second root block imply

$$\begin{gathered}\beta = 51X^2 + 2504, \quad r = \beta(\beta-1)(\beta-2401)(32\beta+343) > 0, \\ (1+149^2 r)R = 149^{2N+1} r, \qquad R_2 = R^2.\end{gathered} \tag{1}$$

In particular $R > 149^3/(1+149^2) > 100$. For the strict lower bound, $r \geq 1$ on the rational input domain, since $\beta \geq 2504$. The retained scalar point is $Q_2 = [a]P$ with $a \neq 0$. Its proof uses the reference point, this scalar block and the scalar pole, not the first or third root block or the aggregate equation being replaced. The same proof gives

$$c = |a| \geq 149^{L_N}, \quad h(\beta) \leq C_3\, 149^{-L_N} c(1 + \log c/3), \quad L_N = \left\lfloor \frac{9(2N-1)}{4} \right\rfloor + 1, \tag{2}$$

where $C_3 = 2^{330770}$. Its uniform completeness statement has one fixed scalar threshold, before the later input-dependent prime choices. These are the exact source scalar assertions retained here.

The fourth root block is also retained: $Q_4 = [b]P$, $b \in \mathbb{Z} \setminus \{0\}$. Nonvanishing of b is essential in the Lucas argument; this block is not removed. The reference point H, the point $H - P$, and their parameters

$$S = y(H)/x(H), \qquad T_H = y(H-P)/x(H-P)$$

are recovered by the unchanged reference equations and scalar argument. They are non-torsion and their indices avoid 0 and 1, as in the source. Neither reference ordinate parameter is zero.

The contact set and exceptional primes remain

$$\Theta = \Theta_0 \cup (2 - \Theta_0), \qquad \Theta_0 = \{0, -1/2, -1, 1/2399, -32/407\},$$

$$\Sigma = \{p : p \leq 76176\} \cup \{2029147\}.$$

For $t \notin \Theta$, let $M(t)$ be the squarefree product of primes outside Σ for which one $v_p(t-\theta)$ is positive and odd. The same full-family modular-height input gives

$$h(t) \leq H_2 M(t)^{7/6} \tag{3}$$

with a fixed constant, including $t = 1$, for which $M(t) = 1$. The proof applies the inherited five-contact bound to both reflected blocks; no new contact family is used here.

3 Reciprocal point pairs, including every exceptional case

For $Q \in E \setminus \{(0,0)\}$ write

$$f(Q) = 1/x(Q), \qquad h(Q) = -y(Q)/x(Q)^2,$$

extended by $f(O) = h(O) = 0$. The reciprocal equation is $h^2 = f - Df^3$. Its coordinates are ordinary affine coordinates after translation by the two-torsion point $(0,0)$, via $(f,h) \mapsto (-Df, -Dh)$.

For each $i = 1, 3$, retain five field coordinates f_i, h_i, g_i, k_i, s_i and use the existing reciprocal-square outputs F_i, G_i. Impose

$$\begin{aligned} F_i = f_i^2, \quad G_i = g_i^2, \qquad h_i^2 = f_i - Df_iF_i, \\ s_i(f_i - g_i) = h_i + k_i, \qquad s_i(h_i - k_i) = 1 - D(F_i + f_ig_i + G_i). \end{aligned} \tag{4}$$

These are five field quadrics. Put, without adjoining outputs,

$$x_i = s_i^2 + D(f_i + g_i), \qquad y_i = Dh_i - s_i(Df_i + x_i). \tag{5}$$

Lemma 3.1. *Every solution of (4) describes an ordered pair $U_i, V_i \in E \setminus \{(0,0)\}$, $U_i \neq V_i$, and their finite difference $Q_i = U_i - V_i = (x_i, y_i)$. Conversely every such ordered pair supplies a unique s_i. Identities, tangent pairs and finite torsion differences are included.*

Proof. Suppress the index and put $r_1 = s(f - g) - h - k$, $r_2 = s(h - k) - 1 + D(f^2 + fg + g^2)$, $r_3 = h^2 - f + Df^3$. Direct polynomial identities are

$$k^2 - g + Dg^3 = r_3 - (f - g)r_2 + (h - k)r_1,$$

$$y^2 - x^3 + Dx = D^2r_3 - D(Df + x)(r_2 - sr_1).$$

They put both reciprocal points and the asserted difference on the curve. If $f \neq g$, the secant through the first translated point and the negative of the second has slope s, and the chord law gives (5). If $f = g$, then $k = -h$. The case $h = 0$ would imply $f = 0$ or $Df^2 = 1$, whereas the last equation demands $1 - 3Df^2 = 0$; this is impossible. Thus $h \neq 0$ and $2sh = 1 - 3Df^2$, exactly the tangent slope, giving the same difference formula. Conversely the chord or tangent law gives the displayed equations. The reciprocal origin $(0, 0)$ represents O on the original curve, so no identity case has been divided out. □

The two ratios for each pair are specified by four additional field quadrics, with $\kappa_1 = 1$, $\kappa_3 = 2$:

$$\begin{aligned} h_iL_i + \kappa_if_iS = 0, \quad (1 - DF_i)L_i + \kappa_ih_iS = 0, \\ k_iL_i^0 + \kappa_ig_iT_H = 0, \quad (1 - DG_i)L_i^0 + \kappa_ik_iT_H = 0. \end{aligned} \tag{6}$$

On the nonidentity domain these give $L_i = \kappa_iz(U_i)/z(H)$ and $L_i' = \sigma L_i^0 = \kappa_i\sigma(z(V_i)/z(H-P))$, with $z = -x/y$. At $U_i = O$ the second equation forces $L_i = 0$, and similarly for V_i. A nonidentity two-torsion numerator has $h_i = 0$, $f_i \neq 0$ and would force $S = 0$, contrary to the retained reference domain. Thus no such numerator supplies a spurious ratio. The difference itself may be finite two-torsion; it is not excluded.

Write

$$\nu(Q_1) = n \in \tfrac{1}{2}\mathbb{Z}, \quad Q_2 = [a]P, \quad 2\nu(Q_3) = t \in \mathbb{Z}, \quad Q_4 = [b]P.$$

The factor two in (6) makes the third recovered label t, not $t/2$. This preserves the integral Lucas equation.

4 Exact transformation of the stored rows

Delete the two seven-field root blocks, old coordinates 155–168 and 183–196, and their respective 18 rational rows 104–121 and 140–157. Append the ten new field coordinates in (4), with the nine field quadrics (4)–(6) for each pair. The count is $212 - 28 + 20 = 204$ rational coordinates and $186 - 36 + 36 = 186$ rational rows.

The reused reciprocal-square slots are $(54, 55), (23, 24)$ for pair 1, and $(58, 59), (27, 28)$ for pair 3. Their second coordinate is one half of the ρ coefficient. Ratio slots are $(31, 32), (33, 34)$

and $(39,40),(41,42)$, respectively. The reference slots are $(19,20),(211,212)$ with ordinary component conventions.

Before compaction make the invertible changes $x_{i,\mathrm{old}} = R^{e_i} x_{i,\mathrm{new}}$, where

$$e_i = 1 \ (39 \le i \le 46 \text{ or } 48 \le i \le 51), \qquad e_i = -1 \ (205 \le i \le 208),$$

and all other e_i vanish. In each affected row collect the total power of R, replace R^2 by its existing slot R_2, and divide by its least occurring power of R. Since $R > 0$, this preserves its zero set. Every remaining power is 0, 1 or 2 and the resulting row is quadratic. This is checked literally for all 22 affected rows; negative least powers are allowed. It unscales the third and fourth label ratios and the two Lucas products, while adjusting the fourth root ordinates inversely. It neither changes the scalar block nor assumes a numerator nonzero.

Replace the old aggregate row 180 by the single rational quadratic

$$8(\lambda - t_1) + 4\,\mathrm{Re}(x_1 + x_3) + \mathrm{Re}\big((C_2 - E_2)^2 + (C_4 - E_4)^2\big) = 0. \tag{7}$$

Here C_2, E_2 are old field slots $(173,174),(175,176)$ and C_4, E_4 are $(201,202),(203,204)$. The retained root identities are $x(Q_j) = (C_j - E_j)^2/(4R^2)$ for $j = 2, 4$. Consequently

$$t_1 = \lambda + T, \qquad T = \tfrac{1}{4}\,\mathrm{Tr}(x(Q_1) + x(Q_3) + R^2(x(Q_2) + x(Q_4))).$$

By [2, Corollary 4.1], $T > 0$, and, for $B = \max(|n|, |a|, |t|, |b|)$, there are fixed constants

$$B^2 \le A(h(R) + h(T)) + J. \tag{8}$$

The weighted height theorem is proved for precisely these enlarged point domains. Reusing the old unweighted theorem here would not be justified.

Keep all contact, local-factor, trace, and pole equations, then compact coordinates in their old increasing order and append the new pairs. Clear every rational row to a primitive integral row with the specified sign convention. Their sum of squares is the unbalanced quartic. The old storage of $(L_1 - \beta + 3)^2$, rather than $(L_1 - \beta)^2$, is retained: its later fourth-power equation performs the correct shift. The constant 2501 in that intermediate row is intentional.

5 Local extraction with half-integral indices

The unchanged local factors, writing q for the contact weight, are

$$1 + q^5 C_2^2 + q^2 D_H^2 + q^4 B_1^4, \qquad 1 + q^5 G^2 + q^2 A_3^2 + q^4 B_0^4,$$
$$1 + q^5 I^4 + q^3 A_2^2 + q^2 A_4^2 + q^4 B_3^4,$$
$$1 + q^5 C_3^2 + q^2 C_1^4 + q^4 B_4^4, \qquad 1 + q^5 C_4^2 + q^2 A_1^2 + q^4 B_2^4.$$

Here $D_H = f(H)$, $A_i = f(U_i)$, $B_i = \sigma f(V_i)$, $B_0 = \sigma f(H - P)$, $C_i = L_i - L_i'$, $I = L_1 - \beta$, and

$$G = L_3^2 - L_2(L_2 - 2)L_3 L_4 + (L_2 - 2)^2 L_4^2 - 1.$$

The factor conventions, including stored affine shifts, are those verified by the exact row transformation.

At a consumed prime, $v_p(q) = -\mu$ with $\mu > 0$ odd. Every finite valuation of a reciprocal abscissa is even: this follows directly from $h^2 = f - Df^3$ when $v(f) \ne 0$, and valuation zero is even too. Zeros may be omitted. The distinct valuation classes in the five factors therefore remain the same as in the source, even for arbitrary points, half-integral indices and zero numerator points. The pole obstruction and reciprocal trace give, at one place above each prime,

$$v(C_2, C_3, C_4, G) \ge 3, \quad v(C_1, I) \ge 1, \quad v(D_H, A_i, B_i) \ge 2.$$

This uses arbitrary finite test valuations, not only positive ones.

For clarity, the formal-group step is also checked on the enlarged domain. Let $\nu(H) = h = m/2$ and $\nu(U) = u \in \frac{1}{2}\mathbb{Z}$. The identities

$$[2m]U = [4u]H, \qquad [2(m-2)]V = [4(u-\nu(Q))](H-P)$$

kill all two-torsion and give exact logarithmic ratios u/h and $(u-\nu(Q))/(h-1)$. At least one of $h, h-1$ is a p-adic unit since p is odd. One ratio is therefore integral. The positive comparison test transfers integrality to the other approximate ratio and then to its exact ratio. Only after this step may the relative formal-logarithm error of order four be used as an absolute congruence. For the third pair multiply both ratios by 2, a unit at these primes. Subtracting the two congruences gives $L_1 \equiv n \pmod{p}$ and $L_3 \equiv t \pmod{p^3}$. The unchanged second and fourth blocks give a, b modulo p^3. Thus for both arguments $t_0 = \lambda, t_1 = \lambda + T$,

$$M(t_j)^3 \mid \Delta, \qquad M(t_j) \mid J_X \quad \text{if } J_X \neq 0, \tag{9}$$

where

$$\Delta = t^2 - a(a-2)tb + (a-2)^2b^2 - 1 \in \mathbb{Z}$$

and J_X is the positive absolute reduced numerator of $\beta - n$. Identity numerator points are covered by $z(O) = 0$ and the second ratio equation. The same argument includes inert places; incompatible formal requirements simply exclude a solution there.

6 One uniform soundness threshold

Elementary height estimates applied to (1) give fixed constants

$$h(R) \leq 4h(\beta) + C_R N + C_0. \tag{10}$$

Indeed the scalar contact polynomial has degree four and its fractional-linear normalization adds at most the height of 149^{2N+1} and a fixed constant. From $c \geq 149^{L_N}$, $c \leq B$, it follows that $N \leq C \log B + C'$. Also (2) implies the uniform coarse estimate $h(\beta) = O(B(1 + \log B))$, independently of N and the solution.

If $M(t_j)^k \mid J_0 \neq 0$ separately for the two arguments, then (3), $h(T) \leq h(t_0) + h(t_1) + \log 2$, and (8)–(10) imply

$$B^2 \leq C|J_0|^{7/(6k)} + C'h(\beta) + C''N + C'''. \tag{11}$$

All constants are fixed before N, X and the witnesses.

If $\Delta \neq 0$, then $|\Delta| \leq 14B^4$ for $B \geq 1$. Take $k = 3$ in (11) to obtain

$$B^2 \leq C_1 B^{14/9} + C_2 B(1 + \log B) + C_3' \log B + C_4.$$

Every term on the right is $o(B^2)$, so B is bounded by a fixed constant. Increasing a single N threshold makes $c \geq 149^{L_N}$ exceed that bound. Hence $\Delta = 0$ for all solutions at all larger N.

The retained b is nonzero. If $t = 0$, the equation $\Delta = 0$ forces $|(a-2)b| = 1$, impossible for $c = |a|$ sufficiently large. Thus a, t, b are all nonzero. The source elementary Lucas argument now applies with integer labels: writing $w = \operatorname{sgn}(a)(a-2)b$ gives $t^2 - ctw + w^2 = 1$. Its positive norm-one solutions are consecutive terms of $U_0 = 0, U_1 = 1, U_{j+1} = cU_j - U_{j-1}$ up to signs and interchange. The divisibility by $a - 2$, reduced modulo $c - 2$ or $c + 2$, forces the relevant index at least $c - 2$. The same source bound is

$$\log B \geq (c-5)\log(c-2) \qquad (c \geq 8). \tag{12}$$

This is why the fourth block, including $b \neq 0$, was retained.

Suppose now $J_X \neq 0$. Writing $\beta = u/v$ reduced and $n = l/2$ gives $|J_X| \leq |2u - lv| \leq 4B \exp h(\beta)$, since $|n| \leq B$ and $B \geq 1$. The extra height terms in (11) are at most $B^2/2$ once B exceeds a fixed constant, by the coarse uniform bounds above. The forced lower bound on c ensures this at all sufficiently large N. With $k = 1$ we obtain

$$B^{5/6} \leq C_5 \exp(7h(\beta)/6), \qquad \log B \leq \tfrac{7}{5} h(\beta) + C_6.$$

Choose N large enough that $(7/5)C_3 149^{-L_N} \leq 1$. This upper bound is at most $c + (c/3)\log c + C_6$, contradicting (12) for all sufficiently large c. Hence $\beta = n \in \frac{1}{2}\mathbb{Z}$. For $X = u/v$ reduced, $v > 0$, this forces $v^2 \mid 102$. As 102 is squarefree, $v = 1$. Soundness holds with one fixed ineffective threshold, not an input-dependent choice of N.

7 Completeness and the half-generator

Fix an integer input and any N beyond the source's fixed scalar completeness threshold. The unchanged scalar preparation gives a positive integer $a \geq 3$, its point $Q_2 = [a]P$, the normalizer R, and a reference progression $h \equiv h_* \pmod{L}$ with $64 \mid h_*, L$, preserving all scalar and finite-place conditions. Choose positive Lucas labels t, b with $\Delta = 0$ as in the source: for $j > 0$ divisible by $a - 2$, take $b = U_j/(a-2)$ and $t = U_{j+1}$. Set

$$Q_1 = [\beta]P, \quad Q_3 = [t]R_0, \quad Q_4 = [b]P.$$

These points are nonzero. The weight R and the new $T > 0$ are now fixed, before any comparison prime is selected.

For an integer reference h, the required points are

$$H = [h]P, \quad U_1 = [h\beta]P, \quad V_1 = [(h-1)\beta]P,$$

$$U_3 = [ht]R_0, \quad V_3 = [(h-1)t]R_0,$$

and the source points for blocks 2 and 4. Their differences are the chosen Q_i. Compact real neighborhoods can be chosen avoiding the finite chart exceptions. There is no rational multiplication map by $1/2$ being used: with $h = 64k$, all displayed points are fixed integer multiples and translates of $[k]R_0$. The source compact-group recurrence, after enlarging the component modulus, therefore supplies joint neighborhoods in this fixed product group. These neighborhoods may depend on the fixed labels, but impose no new lower bound on N.

Use the additive contact coefficient list $\Theta \cup (\Theta - T)$, merge repetitions while retaining their incidences, and clear its finitely many rational denominators by the source square dilation. All coefficients, forbidden primes, congruence moduli and real neighborhoods are fixed before applying the inherited fixed-data simultaneous-prime theorem. It applies to this arbitrary finite rational list, exactly as in the 212 proof.

The selected Gaussian generators already have real part 9 (mod 16) and imaginary part 0 (mod 16). By [2, Lemma 5.1], their old annihilating moduli $\#E(\mathbf{F}_p)/4$ kill R_0 itself in both conjugate groups. Therefore the same relatively prime moduli U, V permit $h \equiv 0 \pmod{U}$, $h \equiv 1 \pmod{V}$ and $h \equiv h_* \pmod{L}$. These congruences make the new third pair formal at its designated places, just as for the doubled generator. This is the additional completeness check required by replacing a tP root by a tR_0 root.

At the selected places the full ratios, with multipliers $1, 2$ on the new pairs, are the exact labels β, t. All comparison errors have valuation at least four, and reciprocal abscissas have valuation at least two. The unchanged factor exponents consequently give unit factors at the chosen place, and reciprocal-trace residue one or two at the other. At other primes all nonselected contact factors contribute square classes as before. At the prepared finite

primes the weight tends to zero; at the two real embeddings it is positive and tends to zero, uniformly on the fixed compact neighborhoods. Thus the actual pole inputs satisfy the source square-denominator, local squareclass, depth and real interval requirements. The retained pole-realization theorem supplies their independent witnesses.

Finally use the reciprocal secant/tangent formulas, the ratio identities, the inverse rescalings by $R \neq 0$, and the retained contact circuits to fill every new coordinate. They satisfy all 186 quadrics. This is a construction of fresh witnesses, not an asserted bijection with every old witness. All input-dependent choices occur after a fixed scalar threshold. Combining this with the fixed soundness threshold proves the quantifier order in Theorem 1.1.

8 Coefficient balancing and complete checks

In the preceding unbalanced rows $A = 149^{2N+1}$. Put $C = 149^N$ and substitute $A = 149C^2$. Rescale the two retained slots R, R_2 by C; in each row divide by the least resulting power of C, then make the integer row primitive. Since $C \neq 0$, this is an invertible rational change at every fixed N. All parameter degrees after division are at most one and all coordinate degrees at most two.

The checker independently reconstructs every deleted, changed and retained row from the pinned 212 source, including coordinate conventions and the two ratio multipliers. It expands each row square by an unordered term-pair sum, rather than importing the producer's expansion. Both presentations have 3046 coordinate monomials; the unbalanced and balanced tables have 3047 and 3048 terms when fixed parameter powers are recorded separately. Every coordinate occurs. An actual fourth-power coefficient of one proves degree four and integer content one.

For the balanced quartic, all coefficient profiles are either linear c_1C or of the form $c_0+c_2C^2$. Exact integers in the checker prove

$$c_1^2 \leq H_0K_0, \qquad |c_0|K_0 + |c_2|H_0 \leq H_0K_0,$$

respectively. Profiles attaining H_0 and K_0C^2 occur. These inequalities imply the exact maximum in Theorem 1.1 for every positive C. The exact comparison $K_0149^{112} < H_0 \leq K_0149^{114}$ proves the threshold 57. Four deliberately corrupted row, ratio, aggregate and quartic data sets are rejected. These are finite algebraic certifications, not a formal proof of the inherited arithmetic statements or an independent non-author review.

References

[1] *A 212-variable quartic with additive height comparisons*, project companion, 13 November 2026, `quartic212_20261113_v25.tex`; exact source tables in `computation/quartic/additive212_v25`.

[2] *Heights of positive weighted elliptic traces*, project companion, 14 November 2026, `weighted_trace_20261114_v27.tex`.

[3] *Trace heights on quadratic twists with $j = 1728$*, project companion, 13 November 2026, `trace_on_all_points_20261113_v25.tex`, Section 2.

[4] Pinned 214-coordinate source and arithmetic applications, `elliotglazer/diophantine-z-q`, commit `1f7e1f1fb0438feda3c9f3da9cc16982126aeeed`, `research/quartic214/proofs/uniform-existence.md`, P03–P20. Numeric source and provenance retained in `computation/quartic/balanced214_v24/source`.